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COMMENTARY

Identifiability and Interpretation of Estimands Under Selection in Perinatal Research

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In this issue of *Paediatric and Perinatal Epidemiology*, Dib and colleagues [1] present simulations investigating the impact of selection due to loss to follow-up or a competing event. Using inverse probability weighting, sibling comparison design and instrumental variable analysis, they demonstrate that under their simulated data distributions, each approach is biased to a similar degree in expectation. They conclude that triangulation across these methods under selection is not informative.

While Dib et al.'s simulations provide important insights under their data-generating assumptions, simulation studies are inherently constrained. Their use of a sharp null hypothesis (no true effect for any individual) enables direct comparison across the three methods, though these methods can target different causal estimands in more general settings. In this commentary, I examine the selection mechanisms they modelled and explore how different estimands might be identified and interpreted. I focus on clarifying when estimands are identifiable and what practical insights they might offer researchers facing selection.

1 | Selection due to Loss to Follow-Up

Dib et al. describe a loss-to-follow-up scenario [1]: Figure 1A (simplified from Figure 1B in [1]) shows a directed acyclic graph (DAG) characterising a study of a vaccination during pregnancy (A) on preterm birth (Y), with loss to follow-up (C). While the authors generated data for up to two pregnancies per individual, for simplicity, I consider a single pregnancy, as both were simulated in the same way, and omit instruments. Dib et al. varied parameters determining loss to follow-up associated with A (v_2) and Y (v_3), as well as baseline risk (v_1) (Figure 1B). While the specific models are unimportant, a value of 0 for a parameter means an edge does not exist.

1.1 | Estimands Under Selection

The average treatment effect (ATE) describes the effect of vaccination among the entire population of pregnant individuals starting the study, whether followed up or not. Under no selection, this is what Dib et al. estimate with inverse-probability weighting. Under the homogeneity assumption (the causal effect of treatment is the same for all individuals in the population) implicit in their data-generating process, the instrumental variable and sibling comparison designs also estimate the ATE. In counterfactual notation, it is defined as:

$$\text{ATE} = E[Y(1)] - E[Y(0)]$$

Another valid causal estimand is the effect of vaccination among those who were not lost to follow-up:

$$\text{ATE}_{C=0} = E[Y(1) \mid C = 0] - E[Y(0) \mid C = 0]$$

Under the sharp null hypothesis of no effect of A on Y for any individual, as in Dib et al.'s simulations, the ATE and $\text{ATE}_{C=0}$ are both 0. Otherwise, the $\text{ATE}_{C=0}$ differs from the ATE when those lost to follow-up have a different distribution of unadjusted causes of the outcome (which will be effect modifiers on some scale; I use the difference scale for simplicity). In certain scenarios, when exposure does not affect follow-up, the $\text{ATE}_{C=0}$

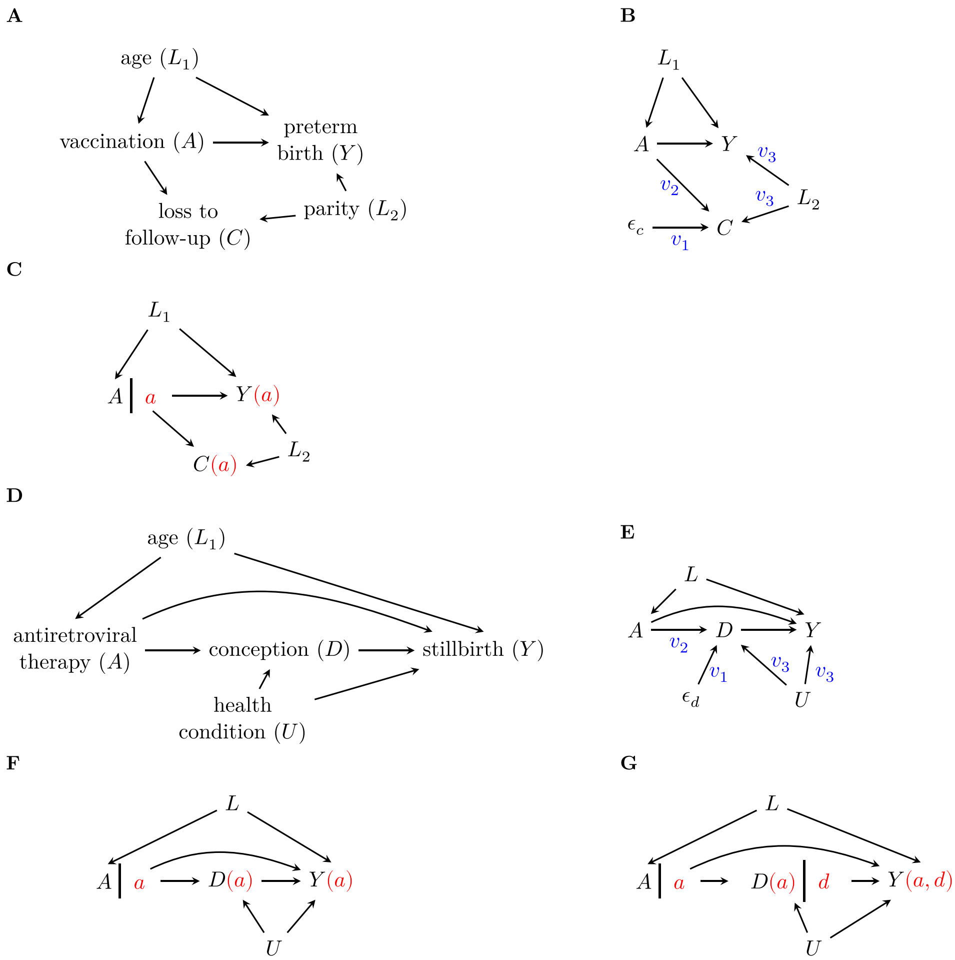

FIGURE 1 | Graphical models for loss to follow-up (A–C) and competing events (D–G). A. Simplified directed acyclic graph (DAG) showing the relationship between vaccination during pregnancy (A) and preterm birth (Y), where loss to follow-up (C) precludes measurement of the outcome among those with $C = 1$. As the exposure–outcome relationship may be influenced by measured confounders like age (L_1), loss to follow-up may also be influenced by various factors. These include exposure itself (perhaps a lack of vaccination leads to more infection, and infected individuals cannot return for follow-up appointments), as well as other factors, possibly related to the outcome (L_2). For example, nulliparous individuals, who have a higher risk of preterm birth, may be more likely to continue participating in a study without other children to care for. B. Parameterised graph showing how loss to follow-up is influenced by exposure (parameter v_2), outcome-related factors (v_3) and baseline risk (v_1 via all other causes ϵ_c). C. Single-world intervention graph for assessing identifiability conditions, where A is set to value a and counterfactual selection status $C(a)$ and outcome $Y(a)$ are depicted. D. Simplified DAG for a study of antiretroviral therapy (A) and stillbirth (Y), where not conceiving or early pregnancy loss (D) is a competing event that prevents observation of the outcome. Unmeasured confounders (U) affect both the competing event and outcome. E. Parameterised graph showing how the competing event D is influenced by exposure (parameter v_2), unmeasured outcome-related factors (v_3) and baseline risk (v_1 via all other causes ϵ_d). F. Single-world intervention graph for an intervention on the exposure only, setting A to value a. G. Single-world intervention graph for the joint intervention setting both A to value a and D to value d.

is identified without adjusting for common causes of selection and the outcome (L_2, i.e., differing effect modifiers), even when the ATE is not identifiable [2].

If exposure does affect selection, then what we can estimate in those scenarios is the net-treatment difference (NTD) [2], which conditions not on the factual sample $C = 0$ but on separate groups defined by counterfactual selection status, where $C(a) = 0$ indicates follow-up under exposure value a:

$$\text{NTD} = E[Y(1) \mid C(1) = 0] - E[Y(0) \mid C(0) = 0]$$

This compares preterm birth among those remaining in the study after vaccination to those remaining after no vaccination, which may be a useful quantity to consider in some settings, but which can differ from 0 under the sharp null hypothesis because different groups are compared.

1.2 | Conditions for Identifiability

Mathur and Shpitser [2] show that two independence assumptions determine which of these quantities are identifiable:

1. $Y(a) \coprod A \mid C(a), W$.

2a. $Y(a) \coprod C(a) \mid W$.

If condition 1 holds for some W, the NTD is identified. This will generally be the case if investigators are careful about measuring $A-Y$ and $A-C$ common causes. If condition 1 holds and A does not affect C, that quantity is equal to $\text{ATE}_{C=0}$. If both conditions hold (regardless of $A-C$ effect), it is also the ATE. Note that these conditions are not directly testable, but are assumptions that rely on correct DAG specification. Mathur and Shpitser show that they can be easily assessed by constructing a single-world intervention graph (Figure 1C), which depicts counterfactuals [2, 3].

In Dib et al.'s data-generating process [1], both conditions hold for $W=\{L_1, L_2\}$, so adjusting for both identifies the ATE. However, the interpretation of the analyses adjusted for only L_1 (their 'unadjusted' results [1]) differs for specific values of the other parameters, and understanding their relationships to identifiability conditions can help us interpret the results. Condition 1 still holds for $W=\{L_1\}$, meaning that without adjustment for factors affecting loss to follow-up, the NTD is identified. When A does not affect C (when $v_2=0$), this is equivalent to $\text{ATE}_{C=0}$ (and there is no bias relative to the ATE in the simulations, as both are 0, which would not necessarily be the case under other $A-Y$ relationships). On the other hand, when either or both of the edges out of C, both with magnitude v_3 in the simulations, do not exist (i.e., there are no common causes of loss to follow-up and preterm birth), then condition 2a holds conditional on L_1, meaning that the ATE can be identified. This interplay of these parameters provides some intuition into the bias expected as v_2 and v_3 vary, whereas the magnitude of v_1 (the baseline risk of loss to follow-up) does not affect identifiability.

2 | Competing Events

Next, Dib et al. consider an event that ensures the outcome cannot occur. The example described alongside their DAG (simplified here in Figure 1D) is a study of antiretroviral therapy (A) in which the outcome is stillbirth (Y), and the competing event is not conceiving or not reaching a gestation of 24 weeks (D). Figure 1E shows how the specified parameters relate to the DAG.

2.1 | Estimands and Identifiability

The estimand that Dib et al. consider is a controlled direct effect (CDE); that is, the exposure–outcome effect if the competing event were entirely prevented. This estimand, involving a joint intervention on both exposure and selection, is unrealistic when the competing event is not generally preventable, but may be more useful in loss-to-follow-up scenarios, where everyone continuing participation may be (at least theoretically) attainable.

$$\text{CDE} = E[Y(A=1, D=0)] - E[Y(A=0, D=0)]$$

Mathur and Shpitser [2] show that the CDE is identified if.

1. $Y(a) \coprod A \mid D(a), W$.

2b. $Y(a,d) \coprod D(a) \mid W$.

Condition 1 is true, as there are no unmeasured confounders of the exposure–outcome or exposure–competing event relationships, which is reflected as an independence in Figure 1F. To check condition 2b we must construct and check the single-world intervention graph for the joint intervention (Figure 1G), where we find it is not true due to unmeasured common causes U of D and Y. Note that condition 2a (to identify the ATE) would never hold for this scenario even without common causes, as there is a direct edge from D to Y, preventing independence of $Y(a)$ and $D(a)$. By instead looking for independence with $Y(a,d)$ on a graph in which the competing event is fixed at a specific value, we could in principle condition on common causes of that event and the outcome to meet condition 2b and instead identify the CDE.

While this implies that we can never identify the ATE in a competing event scenario, we could instead ignore the competing event (by not restricting to $D=0$) and remove D from the DAG, like any ignored mediator of the exposure-outcome relationship. This estimand, referred to as the 'total effect' in the competing events context [4, 5], is the difference in expected outcomes if the population took antiretrovirals compared to if not—that is, the proportion of stillbirths is among all individuals, even if they did not conceive. This estimand includes people not at risk of the outcome in the denominator [6] and conflates the effect of the exposure on the competing event and on the outcome. However, it could be useful in practice in determining resources (e.g., available in-hospital counselling for families experiencing stillbirth could be increased, whether or not antiretroviral therapy has a direct effect on stillbirths, if the absolute number of pregnancies and therefore stillbirths is expected to increase with treatment). The NTD may also be a relevant estimand in the competing events setting for similar reasons [7].

Other estimands may also be of interest in perinatal research (e.g., composite effects, separable effects, principal stratum effects, stochastic direct effects) [5, 7, 8].

3 | Conclusions

While the graphical conditions described by Mathur and Shpitser guarantee identifiability of these estimands under selection, they are not necessary for all causal estimands [2]. For example, odds ratios can recover causal estimands from outcome-dependent selection. Similarly, instrumental variable and sibling comparison designs allow for causal estimation in the presence of unmeasured confounding under other assumptions. However, the latter two methods are not known to have any properties that make them immune to selection bias. In addition, they are sometimes specifically prone to selection bias [9] or poor generalisability [10]. While Dib et al.'s design did not induce additional biases, it did demonstrate that these methods cannot be used to avoid bias either [1].

Instead of attempting triangulation using different methods but the same data, or data with similar selection mechanisms, researchers should seek out data affected by different selection mechanisms for triangulation, while thinking carefully about the estimand for each and whether differences in the magnitude of the estimates reflect different estimands or bias. In any study, researchers should consider what estimand is truly of interest, and if not identifiable under the selection mechanism, whether another informative estimand is. This can depend on the goal of the study: to test the sharp null (i.e., the presence of a causal effect for any person would be a meaningful finding that elucidates some disease aetiology), a number of causal estimands could be meaningful, including the $ATE_{C=0}$ or the CDE. If the results might have practical implications but the specific aetiology of an outcome is less important than a difference in the number of outcomes, an alternative estimand, such as the NTD, may be meaningful.

Author Contributions

L.H.S. wrote the article.

Conflicts of Interest

The author declares no conflicts of interest.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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